AI-Powered Credit Scoring for Nigeria's Unbanked

Concept Note — OPay National Innovation Challenge 2026

Team: Quantro Software Labs	Institution: CUSTECH, Osara	Track: AI & Automation for Social Good	Stage: Concept

The Problem

More than 60 million Nigerian adults cannot get a loan — not because they are irresponsible with money, but because they have no credit history. Banks and lenders only trust people who have used banks before. This leaves out market traders, artisans, farmers, and gig workers who handle money every day but have never had a formal account.

The painful truth is that the data to score these people already exists. Every time someone buys airtime, pays a NEPA token, sends money through OPay, or saves in an ajo group — they are showing financial behaviour. No system captures it. TrustScore does.

The Gap	What It Means
60M+ unbanked adults	More people than the entire population of South Africa
$158B SME financing gap	Small businesses cannot grow without access to credit
Less than 15% bureau coverage	85% of Nigerian adults are invisible to formal lenders
57% of GDP from informal sector	The majority of Nigeria's economy has no credit infrastructure

Our Solution: TrustScore

TrustScore gives every Nigerian a credit score — even those who have never stepped inside a bank. Instead of asking 'have you borrowed before?' we ask 'does your daily financial behaviour show that you can be trusted?' The answer, for millions of Nigerians, is yes.

How It Works — 3 Simple Steps

Step 1 — We collect your financial behaviour data (with your permission)

We look at signals you already produce: how often you buy airtime, whether you pay your electricity bills on time, how active you are on mobile money, and how long you have been contributing to your cooperative or ajo group.

Step 2 — Our AI turns your behaviour into a credit score

A machine learning model analyses over 40 signals from your data and generates a score between 300 and 850. The higher the score, the more creditworthy you appear to lenders. We also tell you exactly why you got the score you did — in plain language.

Step 3 — Lenders use your score to make smarter decisions

Microfinance institutions and cooperative societies connect to TrustScore via a simple API. They type in a user's ID, and within 3 seconds they receive the score, the credit tier (Poor / Fair / Good / Excellent), and a recommendation on how much to lend.

What Data We Use

Data Signal	What It Tells Us
Airtime top-up patterns	How regularly someone earns and spends — a proxy for income consistency
Mobile money activity	How financially active they are and how wide their trust network is
Utility bill payments	Whether they meet obligations consistently — the clearest sign of discipline
Cooperative/ajo records	Months of group savings contributions — the closest thing to loan repayment history
Digital marketplace	Whether they run a business and how active it is

What if someone has no data at all? We give them a 'Thin File' status — not a rejection. We then show them a simple 90-day plan: join a cooperative, pay one bill digitally, use mobile money a few times. After 90 days, they come back and we score them. We do not shut people out — we bring them in.

The Technology Behind TrustScore

We built TrustScore using tools that are reliable, modern, and aligned with the competition's Google partnership. Here is what powers each part of the platform:

Part of TrustScore	Technology Used
AI Scoring Model	XGBoost (industry-standard for financial data) + SHAP for score explanations
API / Backend	FastAPI (Python) — serves scores in under 3 seconds
Database	PostgreSQL on Google Cloud SQL — secure, scalable storage
User Dashboard	React.js — clean web interface for users to view and track their score
Feature Phone Access	Africa's Talking USSD API — dial *123# on any phone, no internet needed
Cloud Infrastructure	Google Cloud Platform — reliable, scalable, aligned with OPay's Google partnership
Data Privacy	AES-256 encryption, user consent controls, full NDPR compliance
Design Workflow	Canva + Google Gemini AI — as required by the competition

Why explainability matters: When your score is 687, TrustScore does not just show you the number. It tells you: *"Your 24 months of cooperative savings added 58 points. Your irregular income reduced it by 34 points. Here is what you can do to improve."* No other Nigerian credit solution does this today.

Who TrustScore Serves	
Who	**Their Situation**
Market traders & artisans	Active every day, cash-based, no bank account. Currently pay exploitative moneylender rates.
Gig & informal workers	No payslip, irregular income, invisible to bureaus. Cannot manage shocks without credit.
Cooperative members	Already proving repayment discipline through ajo — but no system captures it.
Smallholder farmers	Need seasonal credit for seeds and inputs. Cannot access agricultural finance.
Microfinance institutions	Need a faster, cheaper way to assess borrowers beyond slow manual processes.

Why This Matters for Nigeria

Financial exclusion is not just a personal problem for 60 million people — it is a national economic drag. When hardworking Nigerians cannot access credit, they cannot grow their businesses, smooth income shocks, or invest in their families. TrustScore unlocks that trapped potential.

Our Year 1 Targets — Kogi State Pilot

Target	Goal
Nigerians scored	10,000 users in our Kogi State pilot
Lending partners onboarded	5 microfinance institutions and cooperative societies
New credit unlocked	NGN 50 million in loans facilitated through TrustScore scores
Female users	At least 50% — targeting market women who are most excluded
Rural users	30% from non-urban communities
Downstream economic impact	Estimated NGN 115 million — every NGN 1 of credit generates NGN 2.30 in economic activity (IFC research)

How TrustScore Aligns with Nigeria's Goals

Goal	Alignment
CBN Financial Inclusion Strategy	TrustScore directly extends formal financial services to excluded Nigerians
Nigeria Digital Economy Policy	Converts everyday digital behaviour into economic opportunity
SDG 1 — No Poverty	Unlocks credit for low-income Nigerians to invest in their livelihoods
SDG 8 — Economic Growth	Enables informal sector businesses to access growth financing
SDG 10 — Reduced Inequality	Removes the structural barrier that keeps poor Nigerians poor

Why OPay is the perfect partner for TrustScore: OPay already processes millions of mobile money transactions every day across Nigeria. Those transactions are exactly the kind of behavioural signals TrustScore needs. A TrustScore-OPay integration means OPay users get scored automatically — no extra steps. Scored users access credit. Users with credit transact more. Everyone wins.

Our Team

Quantro Software Labs is a team of five 400-level undergraduate students from Confluence University of Science and Technology, Osara. We come from different departments, which means every part of TrustScore has a real expert behind it.

Name	Role	What They Build
Ocheme Emmanuel Onucheojo CGPA: 4.14	Team Lead / ML Engineer	The AI scoring model, feature engineering, and GCP infrastructure
Muhammed Mustapha Ojima-ojo CGPA: 4.64	Data Engineer	The data pipeline, database, and synthetic training dataset
Yahaya Muhammad Tohir CGPA: 4.34	Backend Developer	The API that lenders use to query credit scores
Sadiq Yusuf CGPA: 3.62	Frontend Developer	The user dashboard, lender portal, and Canva-Gemini design asset
Abdulmumin Fatima Onyiekehi CGPA: 3.94	Business Analyst	Market research, business model, and Grand Finale pitch delivery

Team Average CGPA: 4.14 / 5.0 | All members exceed the 3.5/5.0 eligibility requirement.

How TrustScore Makes Money

Revenue Stream	How It Works
API subscriptions (B2B)	Microfinance institutions pay a monthly fee to query TrustScores via our API
Score reports (B2C)	Individual users pay NGN 200–500 to download a credit report to show lenders
White-label licensing	Larger banks license the TrustScore engine to power their own credit products

Our Roadmap

When	What We Do
June–Sept 2026	Build the full working prototype — model, API, dashboard, USSD interface
Oct–Dec 2026	Run Kogi State pilot: 10,000 users, 5 MFI partners, NGN 50M credit target
2027	Scale nationally through OPay's existing 40M+ user base
2028	Expand to Ghana, Kenya, and Tanzania

We are building the infrastructure of trust for Africa's next billion opportunities.

Quantro Software Labs | CUSTECH, Osara | OPay National Innovation Challenge 2026